Departamento de Engenharia Informática
Faculdade de Ciências e Tecnologia
Universidade de Coimbra

IM / ICST

1º Semester / 17 January 2018

NORMAL

2:00 hours – open book

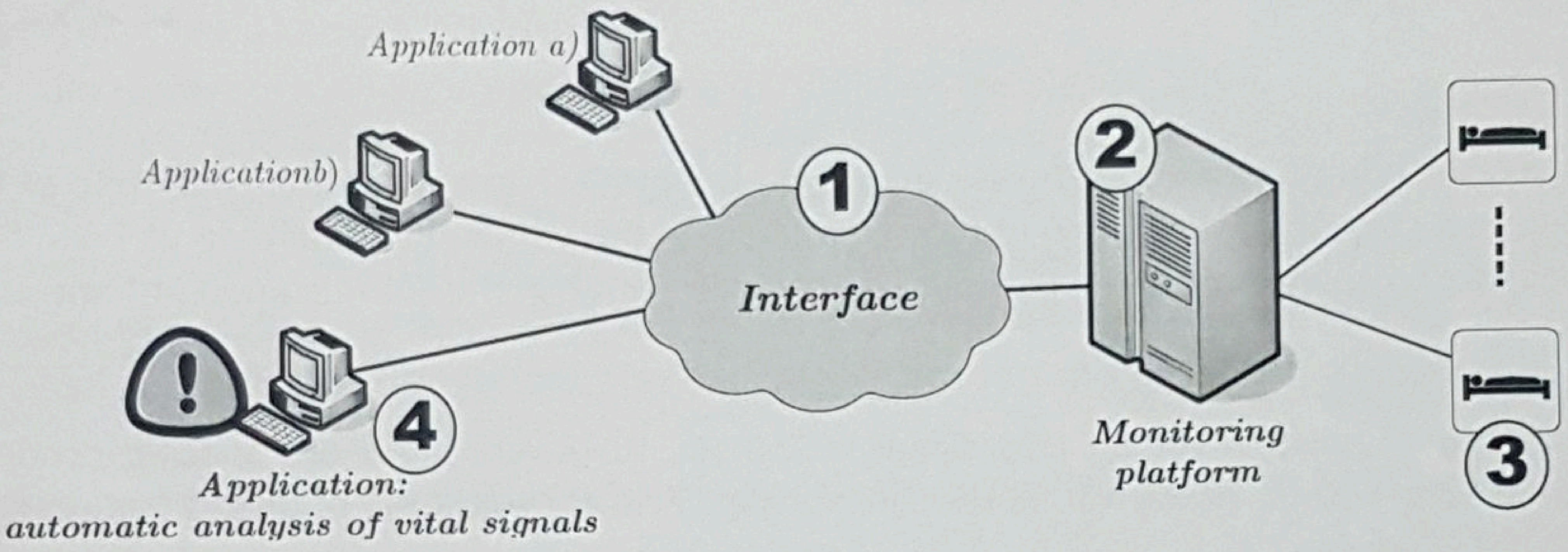
1

1.1

You have been invited to implement a database to support the activities of a dental clinic (appointments, storage of clinical and demographic data). In this context, you have to decide about the possibility for implementing a conventional data base (relational) or for an EAV-Entity Attribute Value data set. Which one would you choose? Justify the reasons for your choice.

1.2

Consider the development of an interface (**1**) to establish the communication between a signal monitoring platform (**2**) that collects ECGs of several patients (**3**) and a set of applications. In particular, the main goal of one of these applications is the automatic analysis (**4**), in order to generate alarms in case a critical condition that requires immediate intervention is detected (e.g. the occurrence of a severe ventricular fibrillation episode).



In this context, do you think that the adoption of the HL7 standard for the implementation of this interface (**1**) is a correct option? Justify your answer.

1.3

Are the HL7 v2.x versions compatible with each other? For example, is it possible to the version v2.2 to communicate with the version v2.5 and vice versa?

1.4

Why is there a significant amount of empty and optional fields in the messages of the standard HL7 version 2.x?

1.5

Consider the following sentence:

"The CDA - Clinical Document Architecture (CDA) is a component of the HL73.0 standard, capable to perform the same role as an Electronic Health Record (EHR)".

Do you agree with this sentence? Justify your answer.

2

2.1

Consider two DICOM systems: system *A* implements SOP X1 and X2 with transfer syntax *Implicit VR Little Endian* and system *B* implements SOPs X1 e X2 transfer syntax with *Explicit VR Little Endian.*

(max. ½ page)

a) Identify the phase of the communication protocol where systems A and B can assess whether they are able communicate with each other or not.

(max. ½ page)

b) Assume you are the engineer responsible to interconnect system A and systems B. Please suggest a least one solution with adequate justification.

2.2

Regarding pHealth systems:

(max. ½ page)

a) Please identify with adequate justification what are the main drivers and gaps for the adoption of these type of systems.

(max. ½ page)

b) Assume that you want to develop a *Continua* compliant pHealth solution that has to integrate vendor-specific data encapsulation protocols for a weight scale and a blood-pressure cuff. Please suggest an adequate architecture for your solution up to the electronic health record. Do identify the main communication technologies to be used.

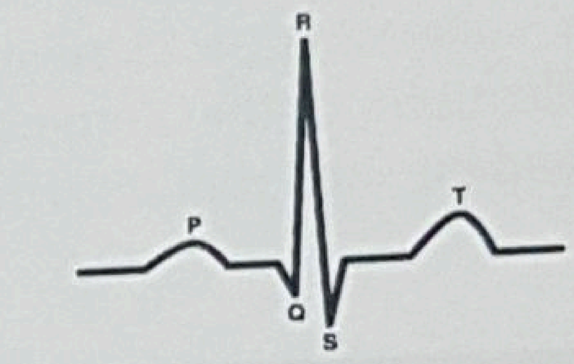
(max. ½ page)

c) Assume that your solution is targeted at the ageing population and that one of the tasks is to timely detect cognitive deterioration through movement patterns (home and mobile contexts). Using the architecture described in b), please detail how you would solve this goal.

3

The Pan & Tompkins algorithm, based on the energy of a signal, addresses the identification of the R peaks of an ECG.

In the analysis of and ECG, besides the QRS complex, two other waves are of particular clinical relevance: the P and the T waves.



3.1

Consider a particular condition where the T wave amplitude is typically higher than the amplitude of the QRS complex (for example, due to a specific configuration of the electrodes). In this condition (amplitude of T wave higher than QRS complex), is it possible to apply directly the Pan and Tompkins algorithm in the identification of T waves? Justify.

3.2

The typical frequency acquisition for an ECG signal is 250 Hz. However, there are specific conditions where this frequency is increased to 500 Hz, or even 1000 Hz. Can you identify some examples of these conditions?

3.3

The Hermite transform is one of the methods commonly used in the detection of PVCs (premature ventricular contractions). Although it follows the same principle as PCA (principal component analysis) - they use base functions in the description of a signal - the Hermite transform uses a different principle.

Can you identify this difference and what are the resulting advantages / disadvantages?

3.4

For the detection of PVC (premature ventricular contractions) time domain methods are generally used. In effect, frequency domain methods are not common to address this issue.

Can you identity the reasons why frequency methods are not used?

Please note that que question is not why time domain methods are used.

3.5

A possible classification for bio-signals is between stationary signal and non-stationary signals. What are the main differences between them?

3.6

Consider the following sentence:

"The parameters calculated from the heart rate variability that characterize a healthy heart are, in general, reduced."

Do you agree with that statement? Justify.